```
# This Python 3 environment comes with many helpful analytics
libraries installed
# It is defined by the kaggle/python Docker image:
https://github.com/kaggle/docker-python
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/"
directory
# For example, running this (by clicking run or pressing Shift+Enter)
will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/)
that gets preserved as output when you create a version using "Save &
Run All"
# You can also write temporary files to /kaggle/temp/, but they won't
be saved outside of the current session

# Use the kagglehub client library to attach Kaggle resources like
competitions, datasets, and models to your session
# Learn more about kagglehub:
https://github.com/Kaggle/kagglehub/blob/main/README.md

import kagglehub
# kagglehub.dataset_download('<owner>/<dataset-slug>')
```

```
/kaggle/input/datasets/parvmodi/cgpa-vs-package-in-lpa/placement.csv
```

Working on a part of dataset

```
df=pd.read_csv("/kaggle/input/datasets/parvmodi/cgpa-vs-package-in-
lpa/placement.csv")
df.head()
```

	cgpa	package
0	6.89	3.26
1	5.12	1.98
2	7.82	3.25
3	7.42	3.67
4	6.94	3.57

Creating 10 datapoints set for visualization

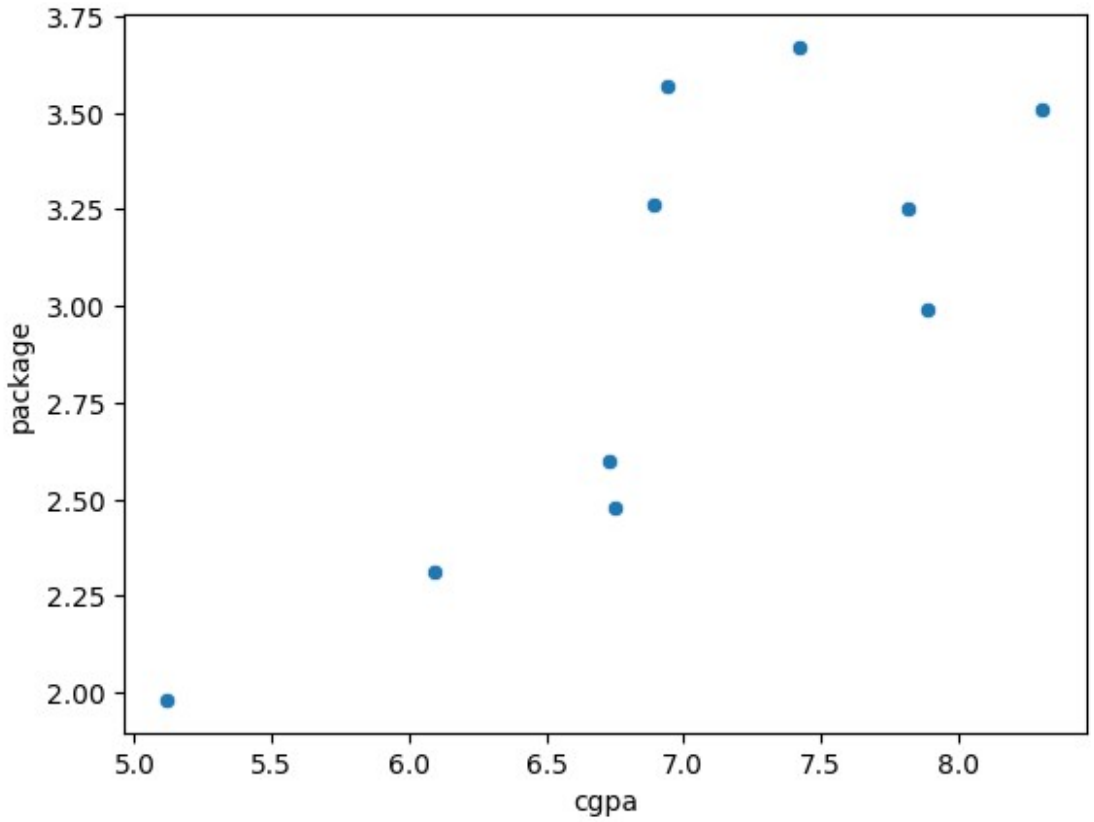
```
df_10=df[:10]
df_10
```

```
   cgpa  package
0  6.89     3.26
1  5.12     1.98
2  7.82     3.25
3  7.42     3.67
4  6.94     3.57
5  7.89     2.99
6  6.73     2.60
7  6.75     2.48
8  6.09     2.31
9  8.31     3.51
```

Impporting Seaborn for visualization of data

```
import seaborn as sns
sns.scatterplot(data=df_10,
                x=df_10["cgpa"],
                y=df_10["package"])
```

```
<Axes: xlabel='cgpa', ylabel='package'>
```



Creating Model

```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
```

Traning Model

```
model.fit(df_10[["cgpa"]],df_10["package"])
```

```
LinearRegression()
```

Lets check the Parameters our model got after training:

Equation y=mx=c

x-> Independent variable(cgpa)

y-> Deependent Variable(Package)

m->Slope/Coefficient

c-> Intercept

```
m=model.coef_
print(f"COEF : {m}")
```

```
COEF : [0.49105006]
```

```
c=model.intercept_
print(f"Intercept : {c}")
```

```
Intercept : -0.4733861893363027
```

Let's try to visualize best fit line

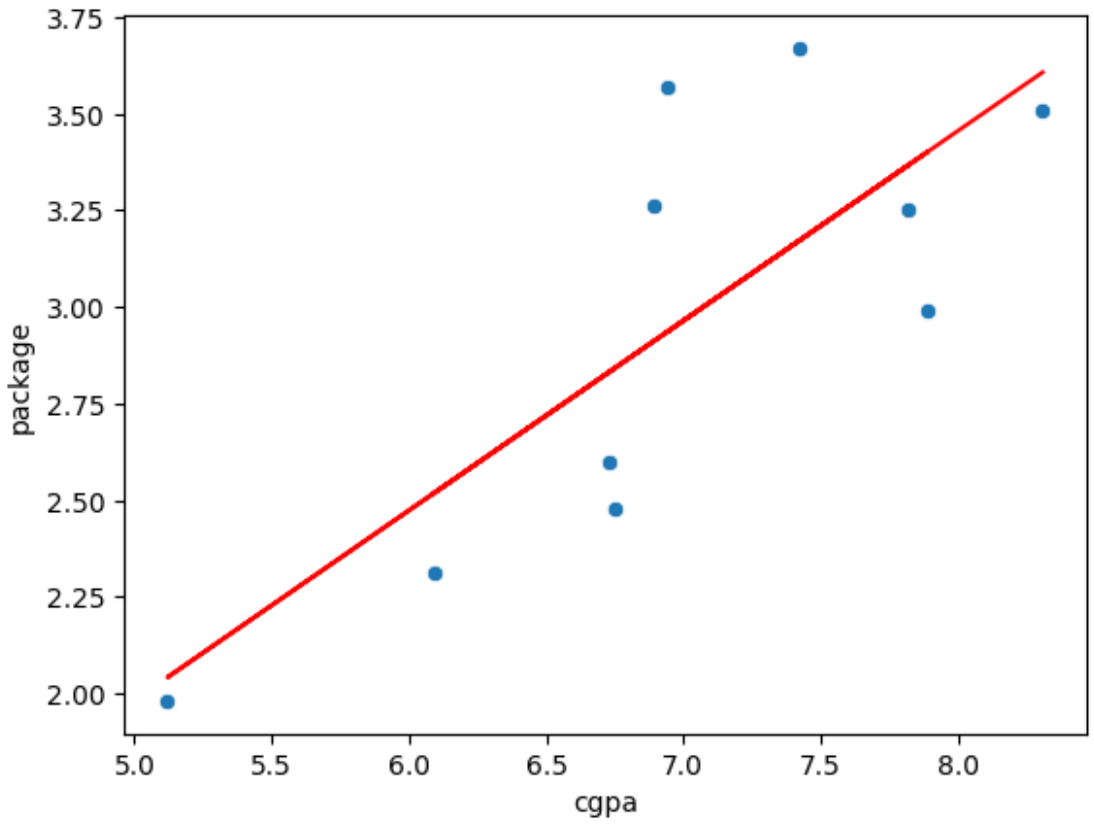
```
import matplotlib.pyplot as plt

X=df_10["cgpa"]
y=df_10["package"]

y_pred=model.predict(df_10[["cgpa"]])

sns.scatterplot(data=df_10, x=X,y=y)
plt.plot(X,y_pred,color="red")
```

```
[<matplotlib.lines.Line2D at 0x7cfae463d1c0>]
```



checking y

```
print(y)
```

```
0    3.26
1    1.98
2    3.25
3    3.67
4    3.57
5    2.99
6    2.60
7    2.48
8    2.31
9    3.51
Name: package, dtype: float64
```

Let's analyze our model with evaulation metrics

```
from sklearn.metrics import mean_absolute_error, mean_squared_error,
root_mean_squared_error,r2_score
```

1. Mean Absoulute Error(MAE)

Manual calculation

```
mae = np.sum(np.abs((y - y_pred)))/len(y)

print(f"MAe calc Manually: {mae}")
```

```
MAe calc Manually: 0.29706897708387314
```

With Scikit-Learn:

```
mae = mean_absolute_error(y,y_pred)

print(f"MAE with sklearn: {mae}")
```

```
MAE with sklearn: 0.29706897708387314
```

2. Mean Squared Error (MSE):

```
mse = np.sum((y - y_pred)**2)/len(y)

print(f"MSE calc Manually: {mse}")
```

```
MSE calc Manually: 0.11987595659200753
```

With Scikit-leran

```
mse = mean_squared_error(y,y_pred)

print(f"MSE with sklearn: {mse}")
```

```
MSE with sklearn: 0.11987595659200753
```

3. Root Mean Squared Error (RMSE) :

Manually:

```
rmse = np.sqrt(mse)

print(f"RMSE manually: {rmse}")
```

```
RMSE manually: 0.3462310739838461
```

With Skicit-learn :

```
rmse = root_mean_squared_error(y,y_pred)

print(f"RMSE with sklearn: {rmse}")
```

```
RMSE with sklearn: 0.3462310739838461
```

4. R2 Score :

Manually :

```
y_mean = np.mean(y)
sse_reg = np.sum((y - y_pred) ** 2)

sse_mean = np.sum((y - y_mean) ** 2)

r2_manual = 1 - (sse_reg / sse_mean)

print(f"R2_Score Manually: {r2_manual}")
```

```
R2_Score Manually: 0.6128737806081344
```

With Skicit-learn :

```
r2_sklearn = r2_score(y,y_pred)

print(f"R2_Score with sklearn: {r2_sklearn}")
```

```
R2_Score with sklearn: 0.6128737806081344
```

Example 02

We will create a data set of our own

```
data={
    "X":[1,2,3,4,5,6,7],
    "y":[1.5,3.8,6.7,9.0,11.2,13.6,16]
}
data
```

```
{'X': [1, 2, 3, 4, 5, 6, 7], 'y': [1.5, 3.8, 6.7, 9.0, 11.2, 13.6,
16]}
```

Converting data into pandas data frame

```
df=pd.DataFrame(data)
df
```

```
   X     y
0  1   1.5
1  2   3.8
2  3   6.7
3  4   9.0
4  5  11.2
5  6  13.6
6  7  16.0
```

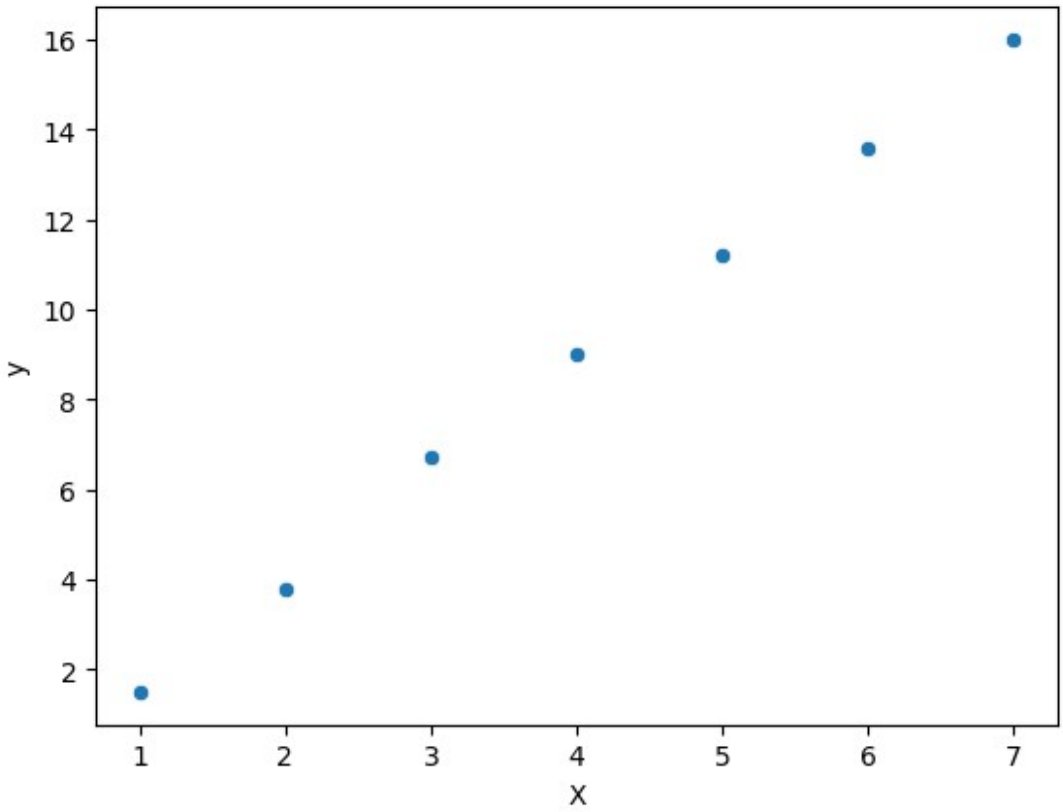
```
df.columns
```

```
Index(['X', 'y'], dtype='object')
```

Let's plot our datapoints

```
sns.scatterplot(
    data=df,
    x=df["X"],
    y=df["y"]
)
```

```
<Axes: xlabel='X', ylabel='y'>
```



Let's train our model

```
model=LinearRegression()

model.fit(df[["X"]],df["y"])
```

```
LinearRegression()
```

```
x=df["X"]
y=df["y"]

y_pred=model.predict(df[["X"]])
```

Testing model with new data points

```
given_cgpa=4

predicted_package=model.predict(np.array([[given_cgpa]]))

print(f"Predicted Package : ",predicted_package[0])
```

```
Predicted Package :  8.82857142857143
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/
validation.py:2739: UserWarning: X does not have valid feature names,
```

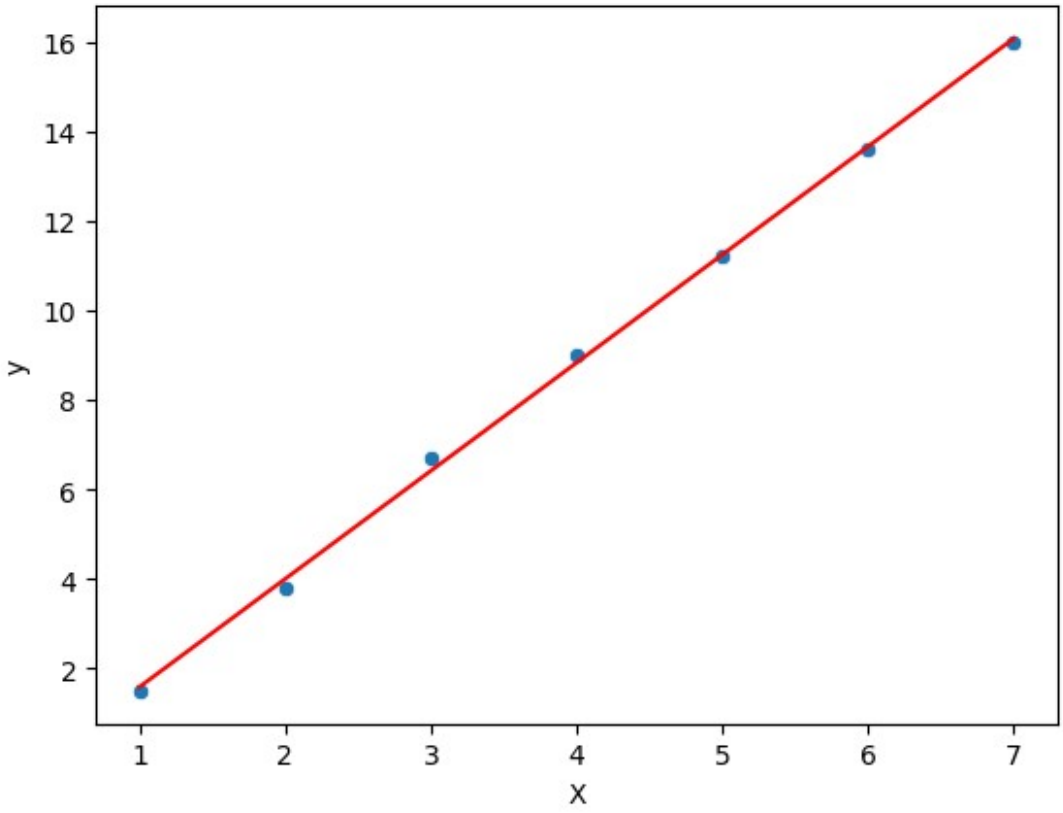
```
but LinearRegression was fitted with feature names
  warnings.warn(
```

Ploting datapoints

```
sns.scatterplot(
    data=df,
    x=df["X"],
    y=df["y"]
)

plt.plot(
    df["X"],
    y_pred,
    color="red"
)
```

```
[<matplotlib.lines.Line2D at 0x7cfae3598050>]
```



Evaulating Model metrics

```
# 1. Mean Absolute Error (MAE):
mae=mean_absolute_error(y,y_pred)
print(f"Mean Absolute Error : {mae}")
```

```
# 2. Mean Squared Error(MSE) :
mse=mean_squared_error(y,y_pred)
print("Mean Squared Error : ", mse)

# 3. Root Mean Squared Error(RMSE):
rmse=np.sqrt(mse)
print("Root mean Squared Error : ",rmse)

# 4. R squared (R2):
r2=r2_score(y,y_pred)
print("R-2 Score : ",r2)
```

```
Mean Absolute Error : 0.13061224489795928
Mean Squared Error :  0.024081632653061243
Root mean Squared Error :  0.15518257844571742
R-2 Score :  0.99896818873402
```

Working on whole data

```
df=pd.read_csv("/kaggle/input/datasets/parvmodi/cgpa-vs-package-in-lpa/placement.csv")
df
```

```
     cgpa  package
0    6.89     3.26
1    5.12     1.98
2    7.82     3.25
3    7.42     3.67
4    6.94     3.57
..    ...      ...
195  6.93     2.46
196  5.89     2.57
197  7.21     3.24
198  7.63     3.96
199  6.22     2.33

[200 rows x 2 columns]
```

```
df.shape
```

```
(200, 2)
```

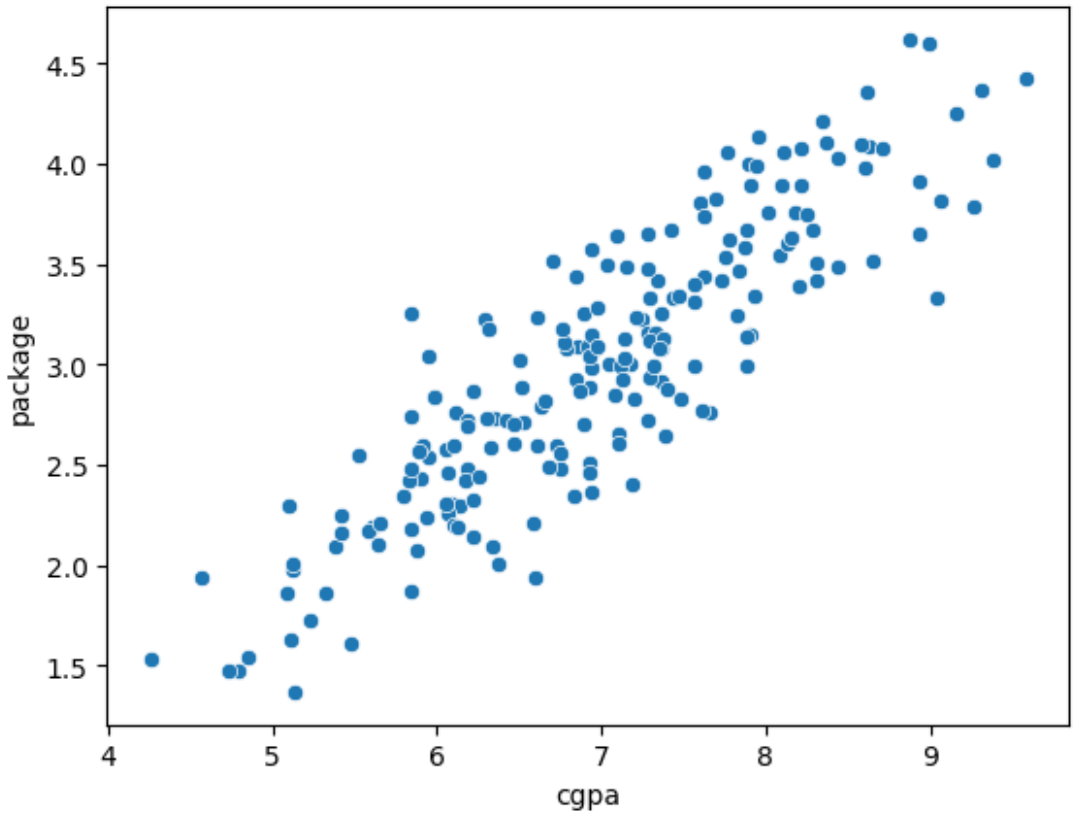
```
df.columns
```

```
Index(['cgpa', 'package'], dtype='object')
```

```
sns.scatterplot(
    data=df,
```

```
    x=df["cgpa"],
    y=df["package"]
)
```

```
<Axes: xlabel='cgpa', ylabel='package'>
```



Using train split method of our data

```
from sklearn.model_selection import train_test_split

X=df[["cgpa"]]
y=df[["package"]]

X_train, X_test, y_train, y_test= train_test_split(X, y,
test_size=0.20, random_state=42)

print(f"X_train Shape: {X_train.shape}")
print(f"X_test Shape: {X_test.shape}")
print(f"y_train Shape: {y_train.shape}")
print(f"y_test Shape: {y_test.shape}")
```

```
X_train Shape: (160, 1)
X_test Shape: (40, 1)
y_train Shape: (160, 1)
y_test Shape: (40, 1)
```

Let's train our model

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()

model.fit(X_train,y_train)
```

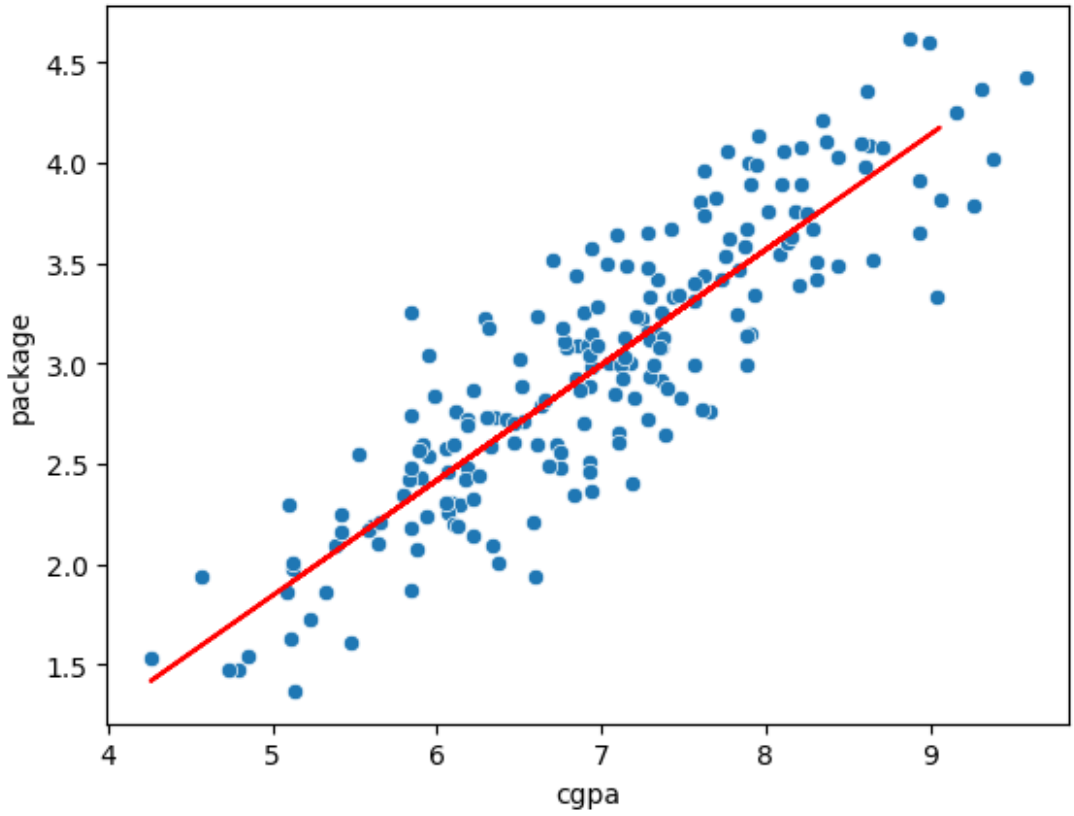
```
LinearRegression()
```

```
y_pred=model.predict(X_test)
```

Let's draw best fit line

```
sns.scatterplot(
    data = df,
    x = df["cgpa"],
    y = df["package"]
)
plt.plot(
    X_test,
    y_pred,
    color="red"
)
```

```
[<matplotlib.lines.Line2D at 0x7cfae358fec0>]
```



Let's evaulate the metrics

```
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score
import numpy as np

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("MAE :", mae)
print("MSE :", mse)
print("RMSE:", rmse)
print("R² Score:", r2)
```

```
MAE : 0.23150985393278373
MSE : 0.08417638361329656
RMSE: 0.2901316659954521
R² Score: 0.7730984312051673
```

Let's Print the parameters our model

```
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
```

```
Intercept: [-1.02700694]
Coefficients: [[0.57425647]]
```